2) $\delta = a - y$

$$a_i = \frac{e^{s_i}}{\sum_{k=1}^{M} e^{s_k}}$$

$$\frac{\partial a_i}{\partial s_j} = a_i(1_{i=j} - a_j)$$

$$A = \frac{\partial a}{\partial s}$$

$$\frac{\partial C}{\partial a_i} = \frac{-y_i}{a_i}$$

so $\nabla_a C = \begin{bmatrix} -y_1/a_1 \\ \vdots \\ y_m/a_m \end{bmatrix}$

$$\delta_j = \frac{\partial C}{\partial s_j} = \sum_{i=1}^{M} \frac{\partial C}{\partial a_i} \frac{\partial a_i}{\partial s_i}$$

$$\Rightarrow \delta_j = \sum_{i=1}^{M} \left(\frac{-y_i}{a_i} \right) a_i (1_{i=j} - a_j)$$

$$\Rightarrow \sum_{i=1}^{M} (-y_i)(1_{i=j} - a_j)$$

$$\delta_j = -\sum_{i=1}^{M} y_i 1_{i=j} + \sum_{i=1}^{M} y_i a_j$$

$\sum y_i 1_{i=j} = y_j$ and a_j is constant

$$\Rightarrow \delta_j = -y_j + a_j \sum_{i=1}^{M} y_i$$

<u>so</u>: $\delta_j = -y_j + a_j \quad [\sum_i y_i = 1]$

$$\Rightarrow \delta = a - y$$